

Math 306 Exercise Set 6

Textbook exercises from Section 6.A: 4, 6, 7, 23, 24.

- 1.** Define a square root of $T \in \mathcal{L}(V)$ to be a linear map $S \in \mathcal{L}(V)$ such that $S^2 = T$.
 - a. Show that the $T \in \mathcal{L}(\mathbb{C}^2)$ defined by $T(a, b) = (-a + 2b, -2a + 3b)$ has a square root $S \in \mathcal{L}(\mathbb{C}^2)$ defined by $S(a, b) = (-b, a - 2b)$.
 - b. Give an example of a $T \in \mathcal{L}(\mathbb{C}^2)$ that has two different square roots.
 - c. Let S be a square root of T with eigenvalue λ . Prove that if λ is an eigenvalue of S , then λ^2 is an eigenvalue of T .
 - d. Prove that there is no square root for the $T \in \mathcal{L}(\mathbb{C}^2)$ defined by $T(a, b) = (0, -a)$.
- 2.** Give an example of a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (the underlying field is $\mathbb{R}$, not $\mathbb{C}$) for which there is no basis of $\mathbb{R}^2$ such that the matrix for T is upper triangular.
- 3.** Proof or counterexample: Take $S, T \in \mathcal{L}(V)$. If λ and μ are eigenvalues of S and T , then $\lambda + \mu$ is an eigenvalue of $S + T$.
- 4.** Define trace A to be the sum of the diagonal entries in the square matrix A . Let $M_{m,n}(\mathbb{R})$ be the vector space of $m \times n$ matrices with entries in $\mathbb{R}$ and define $\langle A, B \rangle = \text{trace}(B^\top A)$.
 1. Show that $\langle A, B \rangle$ is an inner product on $M_{m,n}(\mathbb{R})$.
 2. Show that if $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are written as $n \times 1$ matrices, then $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y}$ where $\cdot$ denotes the dot product.
 3. Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Find $\|A\|$ and the distance from A to the 3×3 identity matrix.
 4. Let A be an $n \times n$ matrix. This exercise shows there is an invertible matrix B arbitrarily close to A . Let $\varepsilon > 0$ and take $\delta \in [0, \varepsilon/\sqrt{n})$ such that δ is not an eigenvalue of A . Show that $B = A - \delta I$ is an invertible matrix such that $\|A - B\| \leq \varepsilon$.